AUTOMATED PACKAGE SEPARATOR

Prepared for: Sinai Health Systems
Team 181
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The current solution is manually separating pills

Alba Munoz Sevillano



Gap

Continuous Strain & time consuming

Need

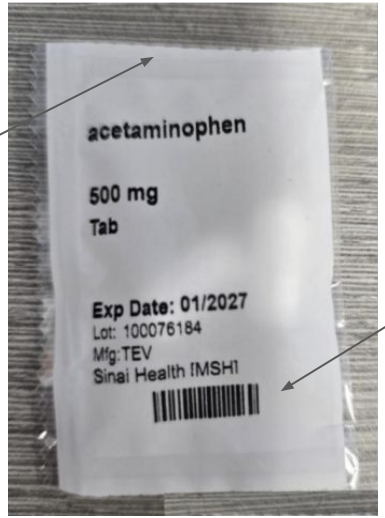
An autonomous technology to efficiently separate pills

Scope

Mount Sinai & Hendrick Bridgepoint Hospital Pharmacies



Package Integrity



Barcode Readability

What are the impacts?

Alex
Yu



1) Occupational Strain

- Thousands of tears per day [2]
- Long term health risk

2) Human Capital Waste

- 1-3 years of medical experience [3]
- 2 year college accreditation [4]

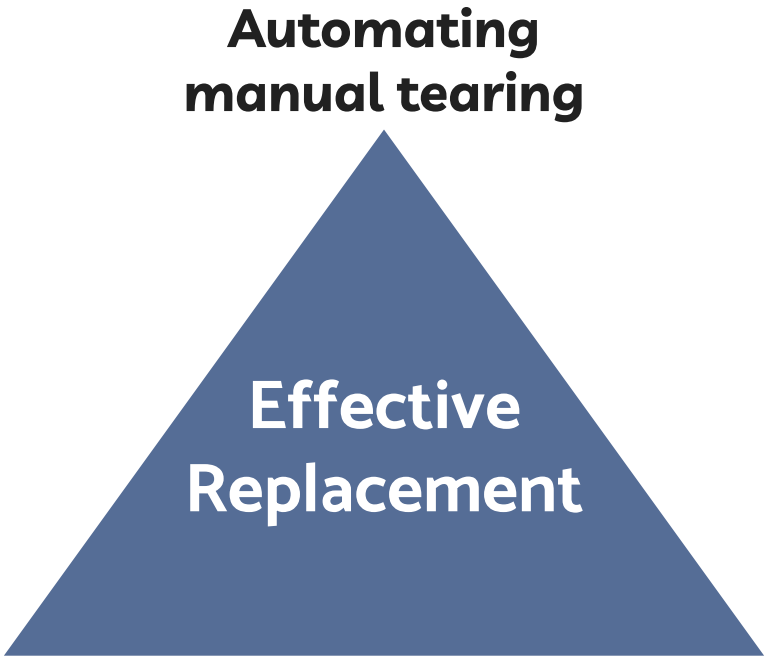
3) System-Wide Impact

- Affects over 50% of hospitals in Canada [2]
- Diverts technicians from more important tasks [3]

PharmaSlice effectively replaces manual tearing

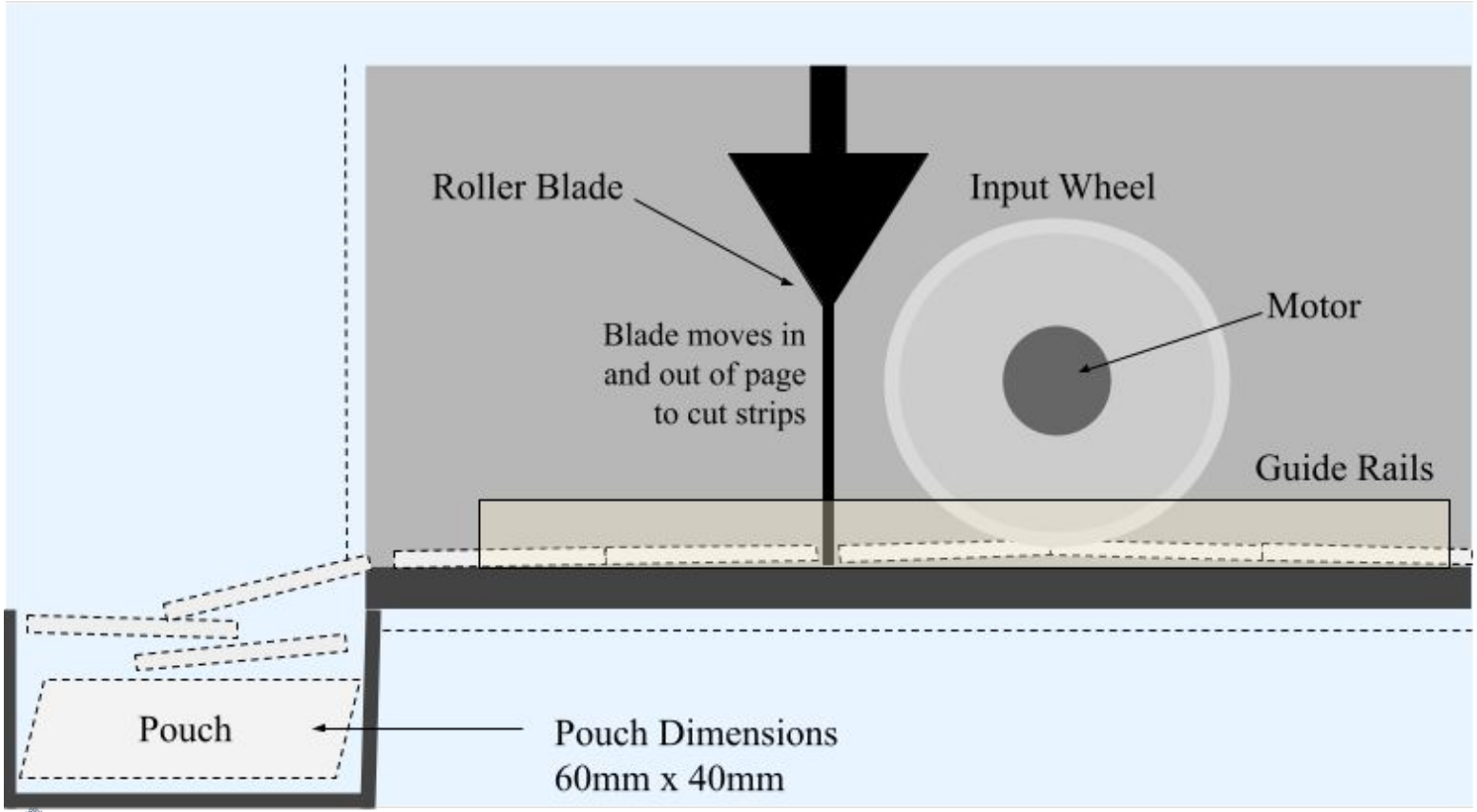
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Our design

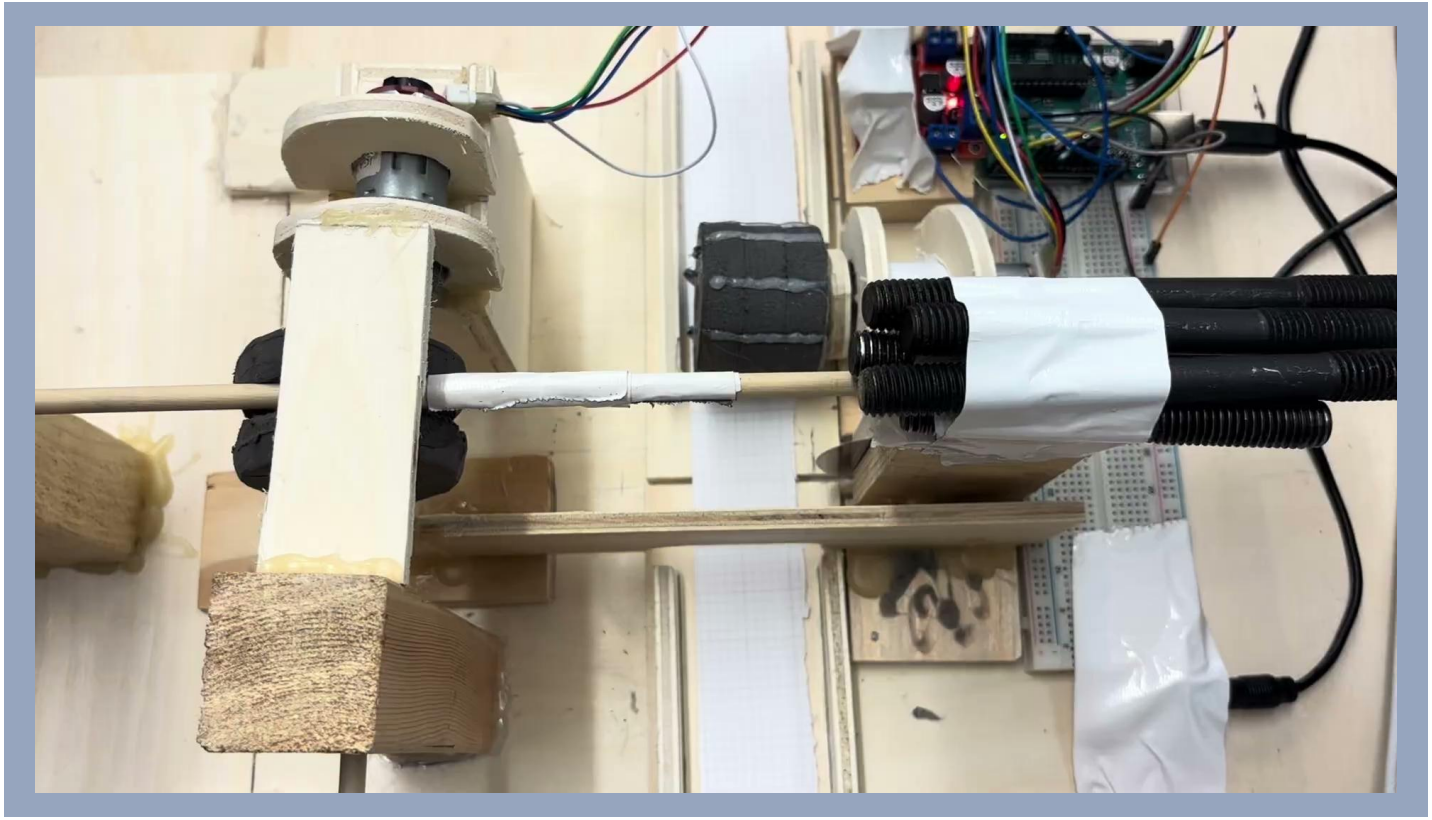
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An automated solution

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PharmaSlice meets all critical constraints

Boen Peng



1 VARIABLE CUTTING INTERVAL

2, 5, 10 individual pill packages per strip

2 CUTTING ACCURACY

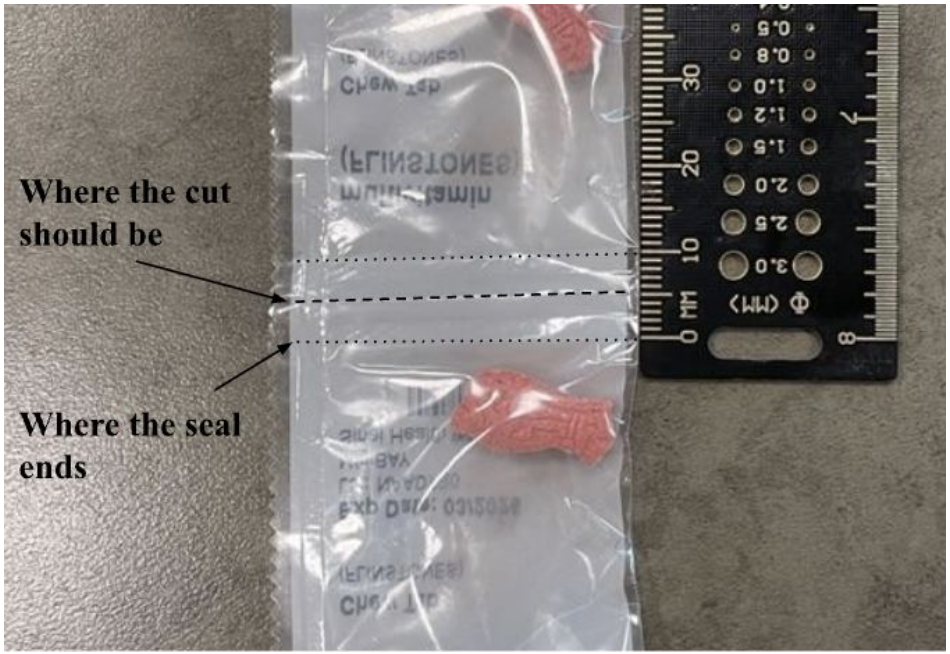
Maximum error of ±3mm to prevent leaks

3 SAFETY/SHARP PARTS

0 cm of exposed moving parts

4 PILL COMPATIBILITY

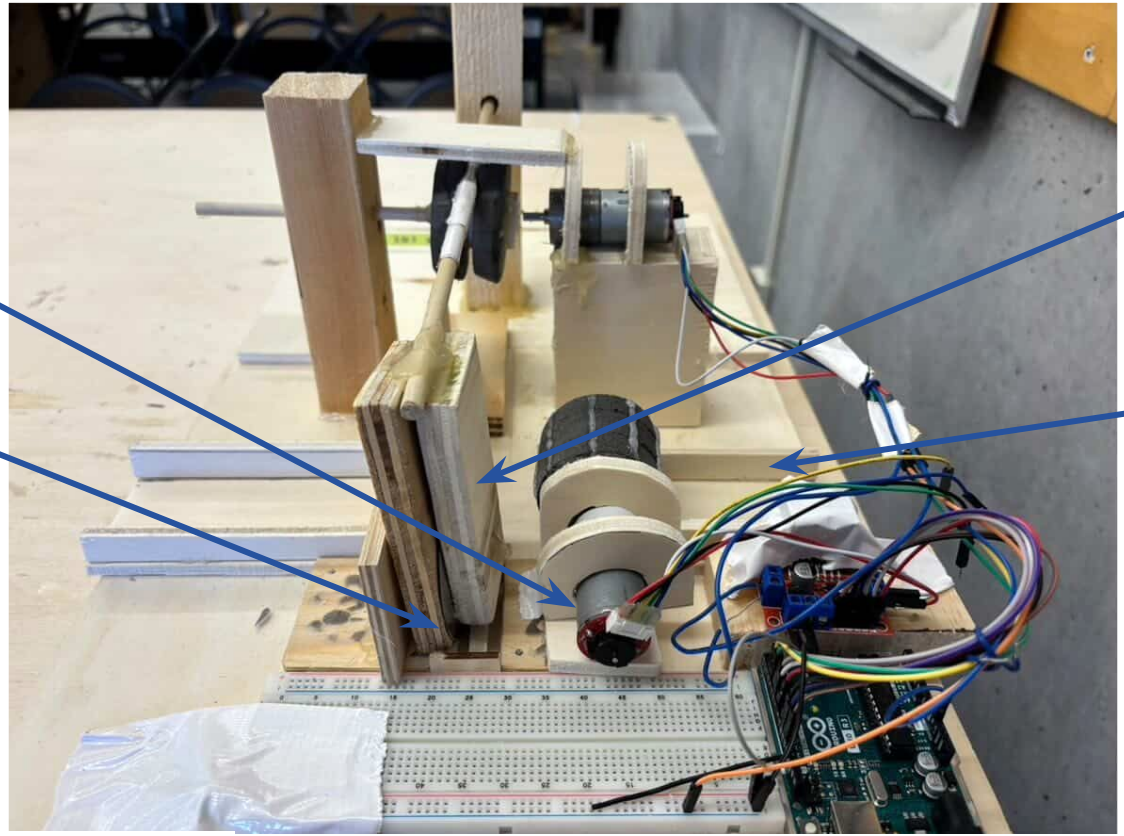
Accommodates pill sizes from 6~22 mm [5][6]



PharmaSlice meets all critical constraints



Boen Peng



C.1: Cutting Interval

C.2: Cutting Accuracy

C.3: Safety

C.4: Pill Compatibility

PharmaSlice satisfies key design objectives

James Lee



1 HUMAN INTERACTION

< 60 min per day machine interaction

2 TOTAL PRODUCTION

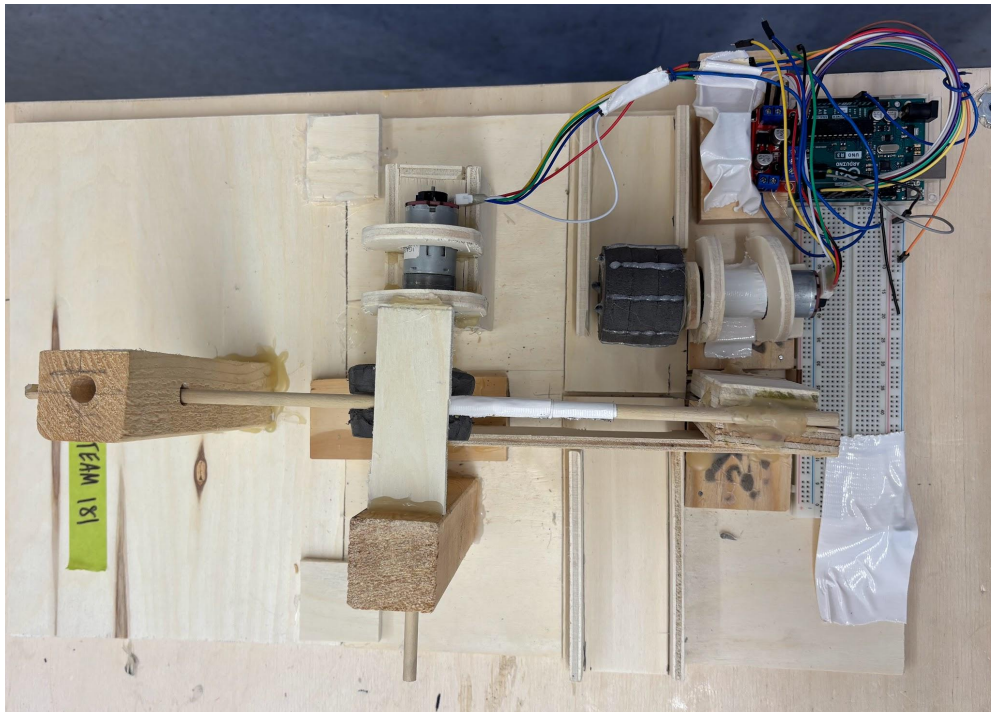
≥ 1750 pouches cut per hour

3 SUCCESS RATE

≥99.9% of packages undamaged

4 LOADING CAPACITY

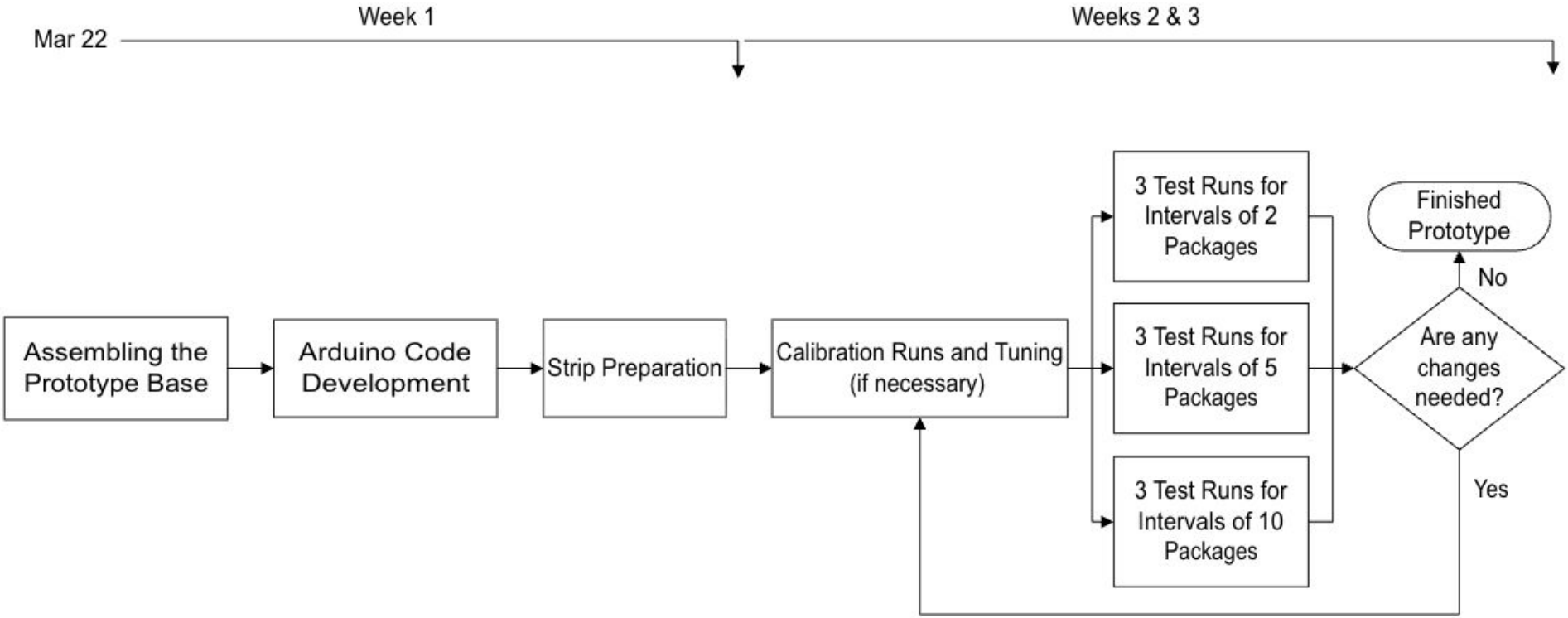
Load ≥500 packs at a time



Project Timeline

James Lee





Testing Results

Alba Munoz Sevillano



Limitations:

Low-fidelity prototype	Materials and performance degrade under repetitive use
Quick build tolerances	Alignment issues & reduced cutting accuracy
Unrealistic testing environment	Unrepresentative of chaotic pharmacy environment

Results:

0.1: Human Interaction:

- <30s per strip
- <40 min / day

0.2: Total Production (per hr):

- Strips of 2 - **2,162/h**
- Strips of 5 - **3,750/h**
- Strips of 10 - **4,960/h**

What is next for PharmaSlice?

Nolan Cheung



Upgrade Materials

Shift to CNC manufacturing for industrial-grade durability and part consistency.

Reduce Tolerances

Tighten tolerances to eliminate alignment errors and increase cutting accuracy.

Design Output Mechanism

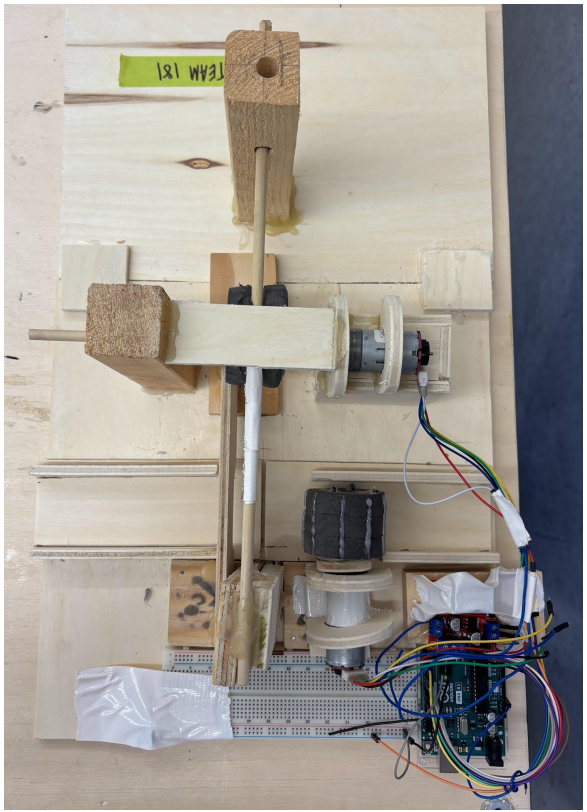
Integrated system to automatically clear cut pouches from the workspace.

Jam Detection

Real-time sensor monitoring to prevent hardware damage and minimize operational downtime.

Conclusion

Nolan
Cheung



1. Automates Pharmacy Process

- Automates high-frequency manual motions to prevent unnecessary workplace strain injuries

2. Improves Pharmacy Efficiency

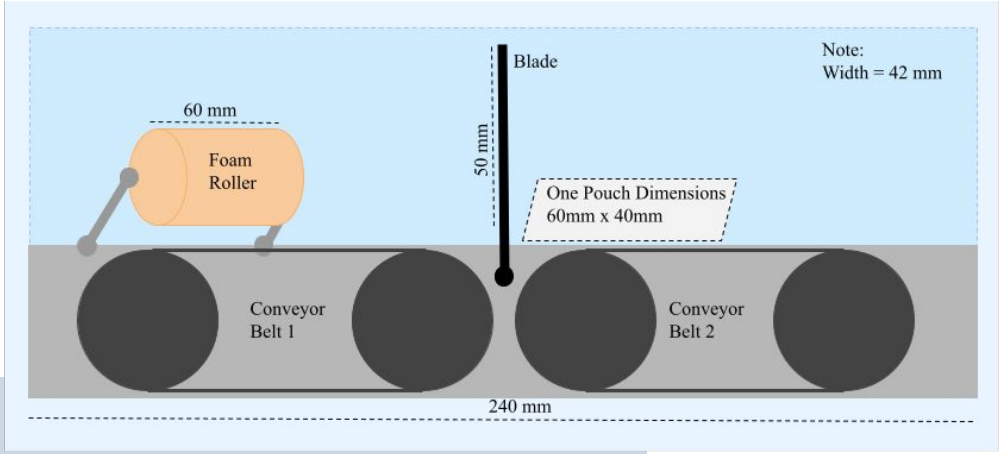
- hrs -> 40 min / day
- Reinvest human resources on more demanding tasks

3. Wide Scale Impact

- Simple architecture capable of supporting diverse pharmacy workflows across North America

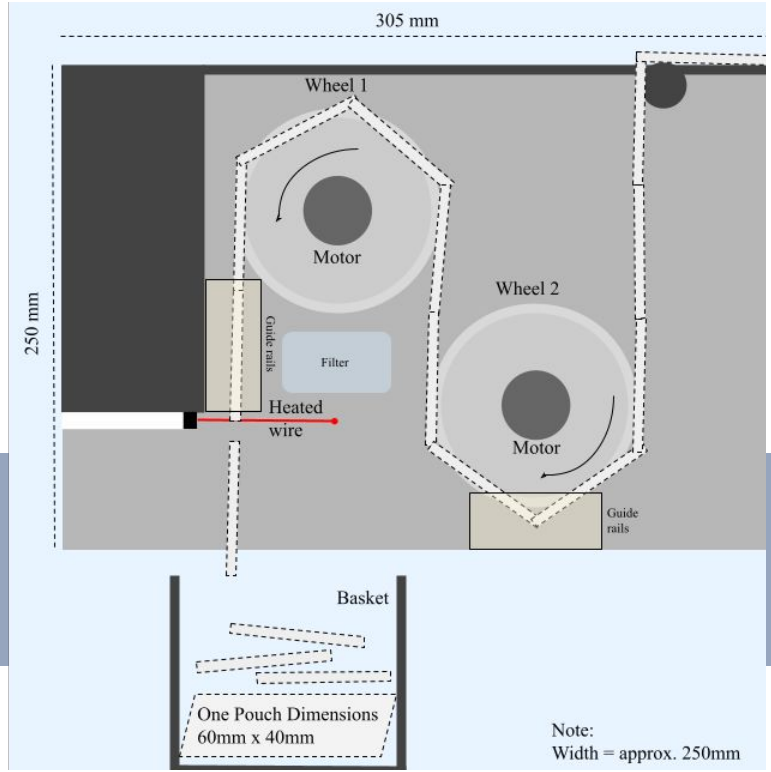
APPENDIX

Alternative Designs



SliceFlow

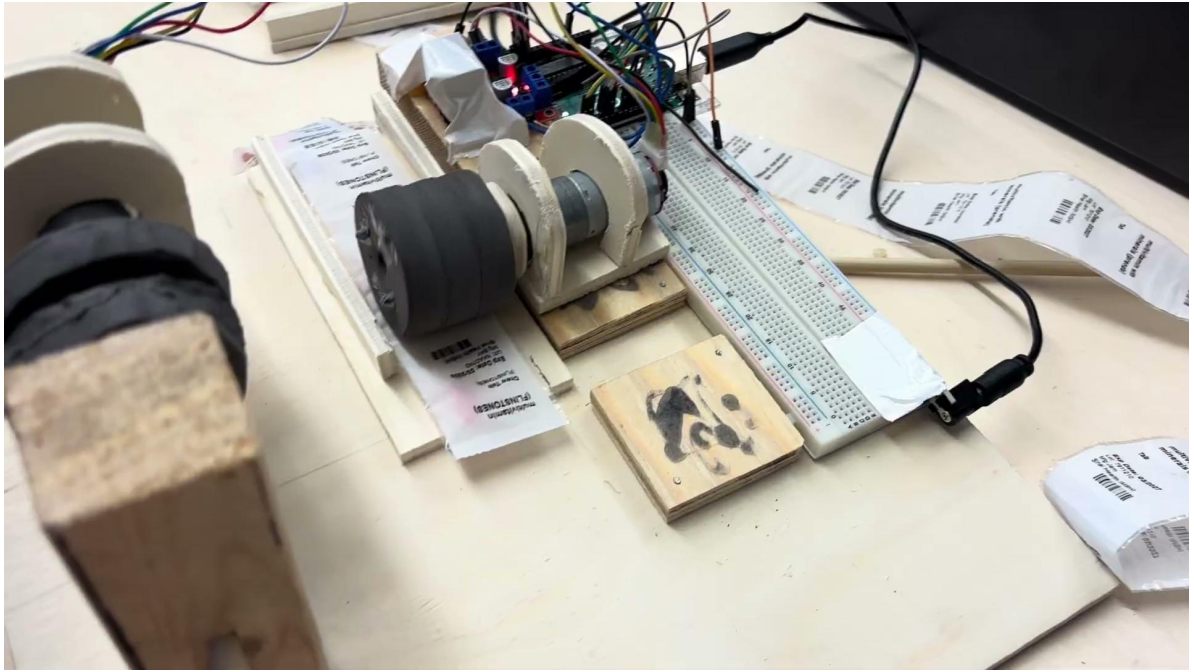
Thermocut



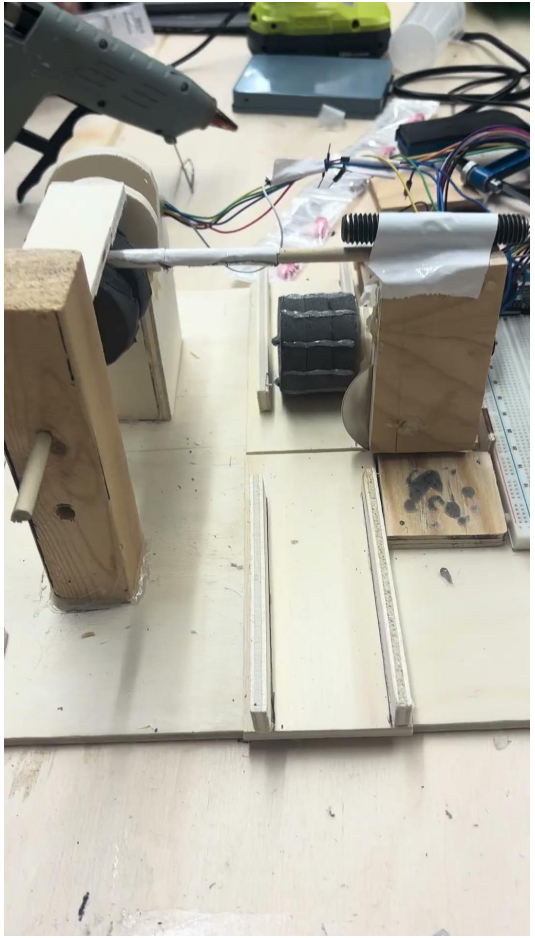
Pugh Chart

Objective	Design 1 (PharmaSlice /Datum)	Design 2 (ThermoCut)	Design 3 (SliceFlow)
O.1 Human Interaction	S	-2	0
O.2 Total Production	S	0	0
O.3 Success Rate	S	0	0
O.4 Loading Capacity	S	0	0
O.5 Maintenance Frequency	S	0*	-1
O.6 Internal Visibility from Outside	S	0	0
O.7 Dimensions	S	+1	0
O.8 Spending	S	-1	0
Total	**0**	**-2**	**-1**

More videos of the prototype



Roller motion tested with pills inside the package



Roller motion and cutting motion synchronization testing without blade attachment or packed pills input

Calculations

Advancement Time

Batch of 2: 490 ms x 2 = 980 ms
Batch of 5: 490 ms x 5 = 2,450 ms
Batch of 10: 490 ms x 10 = 4,900 ms

Total Cycle Time

Batch of 2: 980 ms + 2,000 ms + 350 ms = 3,330 ms
Batch of 5: 2,450 ms + 2,000 ms + 350 ms = 4,800 ms
Batch of 10: 4,900 ms + 2,000 ms + 350 ms = 7,250 ms

Cycles per Hour

Batch of 2: 3,600 ÷ 3.33 ≈ 1,081 cycles/hour
Batch of 5: 3,600 ÷ 4.80 = 750 cycles/hour
Batch of 10: 3,600 ÷ 7.25 ≈ 496 cycles/hour

Max Strips/Packets per Hour

Batch of 2: 1,081 strips OR 2,162 packets
Batch of 5: 750 strips OR 3,750 packets
Batch of 10: 496 strips OR 4,960 packets

References

[1] "Mount Sinai Hospital redevelopment project," image, *REMI Network*, Aug. 1, 2024. Accessed: Apr. 8, 2026. [Online]. Available: Mount Sinai Hospital completes historic redevelopment project.

[2] Sinai Health, "Automated Package Separator Project (APS112)", Toronto, Ontario, 2025.https://docs.google.com/document/d/1UySmorw7jF-qF4XKM2XXR8DfcHE9iK1zTY0pORAH0sQ/edit?usp=sharing

[3] "Pharmacy Technician, Pharmacy," Mount Sinai Hospital (Toronto), Sinai Health, *LinkedIn Jobs*. Accessed: Apr. 8, 2026 [Online]. Available: https://www.linkedin.com/jobs/view/pharmacy-technician-pharmacy-at-mount-sinai-hospital-toronto-sinai-health-4380812388/

[4] "Pharmacy technician," *Centennial College*. Accessed: Apr. 8, 2026. [Online]. Available: https://www.centennialcollege.ca/programs-courses/full-time/pharmacy-technician.

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